1 Motivation

- Growing digital generators face rising inference energy, memory and latency costs.
- Free-space optics performs large parallel linear transformations at light speed and near-zero arithmetic energy.
- The goal is a reconfigurable optical generator competitive with digital neural networks.

2 Snapshot model

NOISE TO PHASE TO IMAGE

- A three-layer digital encoder maps Gaussian noise and class information into a 2D phase pattern.
- An SLM displays this optical seed; coherent light passes through an 800 x 800 learnable phase decoder.
- A sensor captures the output intensity as a novel monochrome or color image in one optical snapshot.

3 Training

- First train and freeze a proxy DDPM that supplies paired noise and target samples.
- Jointly optimize the digital phase encoder and diffractive decoder by differentiable wave propagation.
- Iterative variants mimic diffusion and can learn self-supervised; multi-color systems synthesize faces and butterflies.

OPTICAL GENERATION

Optical Generative Models

Diffusion-Inspired Image Synthesis with Reconfigurable Optics

Encode random noise into optical phase seeds with a shallow digital network, then let a jointly optimized diffractive decoder synthesize new images through physical light propagation.

photonics · diffractive optics · generation

4 Physical design

- A new dataset requires updating the decoder phase pattern, not changing the optical architecture.
- Output diffraction efficiency can enter the loss; deeper decoders improve quality at a fixed efficiency.
- Precomputed phase seeds make local optical inference delay and compute energy minimal.

5 Evidence

DIGITS TO FACES

- Demonstrations cover MNIST, Fashion-MNIST, Butterflies-100 and CelebA, including visible-light hardware.
- More than 94.2% of generated MNIST digits are recognized; IS/FID support diversity and fidelity.
- Snapshot generation matches a deeper nine-layer digital MLP while requiring about 3x fewer digital FLOPs.

6 Potential & limits

PHYSICAL AI TRADE-OFFS

- Spatial/spectral multiplexing could generate many images in parallel, including 3D or on-chip variants.
- Misalignment, fabrication error, limited phase bit depth and sensor noise can degrade the physical model.
- The shallow encoder and illumination still consume resources; current systems remain specialized image generators.